

Open Source Cellular Solutions vs. Hosted Cellular Systems: A Comparative Technical Review

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Abstract

This article presents a comparative technical evaluation of Open Source Cellular Solutions and Hosted Cellular Systems as alternative approaches to enterprise telephony deployment. Three principal open source software packages are reviewed: OpenBTS, OpenBSC, and Asterisk. The architectural principles underlying each platform are examined alongside their respective integration characteristics and deployment contexts. The article then evaluates Hosted Cellular (VoIP) Systems, covering the operational model by which service providers manage network infrastructure on behalf of enterprise clients. The benefits and weaknesses of both approaches are systematically assessed across dimensions including cost, customisation, security, interoperability, technical complexity, and scalability. The analysis concludes with a recommendation framework to assist organisations in selecting the appropriate solution based on budget constraints, technical capability, and operational requirements.

1. Introduction

In contemporary telecommunications environments, low-cost open source systems can be integrated with existing IP infrastructure to deliver telephony services to a geographical area or to an enterprise (Lambrinos, 2008). These emerging solutions are progressively displacing traditional circuit-switched telephone systems with IP telephony architectures, most notably Voice over IP (VoIP).

Open source deployment of GSM network components has gained considerable traction in recent years. Platforms such as OpenBTS, OpenBSC, and Asterisk have created viable alternatives to proprietary cellular infrastructure. Alongside these, hosted telephony services have emerged as a compelling option for small and medium-sized enterprises (SMEs) that require access to current-generation telephone infrastructure without the capital expenditure of owning or operating those systems. In a hosted model, telephone facilities and equipment are managed by the cellular service provider, enabling users to connect from any location, including home offices, hotels, or mobile devices, while remaining integrated with the organisation's internal telephone system.

This article examines open source cellular systems and hosted cellular systems in parallel, discusses their defining architectural and operational characteristics, and provides a structured basis for deployment decision-making.

2. Open Source Cellular Solutions

2.1 Overview

Open source refers to software whose source code is made available without restriction for use, modification, and redistribution by any party. In the open source model, development contributions and fixes are distributed freely, typically managed by communities of developers, volunteers, and dedicated forums organised around the software project.

Open source solutions are widespread in the cellular network industry, with notable projects including OpenBTS, OpenBSC, and Asterisk operating as fully functional GSM and VoIP infrastructure components. These platforms have attracted adoption in research institutions, academic environments, and cost-constrained enterprise deployments.

2.2 Open Source Cellular Software Platforms

A. OpenBTS

OpenBTS deploys a standalone Base Transceiver Station (BTS) that bypasses the traditional Base Station Controller (BSC) and Mobile Switching Centre (MSC) components. Instead, it uses Session Initiation Protocol (SIP) to connect to a software-based telephone switch (Tsou et al., 2012). Through this architecture, OpenBTS routes mobile and external connections over a Voice over IP protocol, functioning as a GSM access point that operates independently without requiring signalling and switching from a conventional core network.

The radio air interface software supports multiple hardware options, with the Universal Software Radio Peripheral (USRP) being the most commonly deployed. By integrating MSC and BSC functions within a single software switching system, OpenBTS handles all switching and calling operations internally.

A secondary function of OpenBTS is bidirectional conversion between GSM data and VoIP data packets. When an OpenBTS node receives GSM signals, it demodulates those signals and translates them into VoIP packets for onward transmission. Call initiation occurs once signalling between the two call parties is complete (Range Networks, 2012). This signalling is managed by the PBX (Private Branch Exchange) component, which operates using SIP.

B. OpenBSC

OpenBSC adopts a more traditional approach to cellular network implementation compared to OpenBTS. Whereas OpenBTS creates an alternative GSM architecture, OpenBSC operates on the network side of a standard GSM network architecture (Welte et al., 2012). As a result, OpenBTS and OpenBSC may be deployed together within a cellular network architecture.

OpenBSC is compatible with one or more commercially available BTS components and is capable of constructing a standalone GSM network that consolidates the functions of a BSC (Base Station Controller), AuC (Authentication Centre), MSC (Mobile Switching Centre), EIR (Equipment Identity Register), VLR (Visitor Location Register), and HLR (Home Location Register) into a single integrated platform (Baharlooee and Hashemi, 2009).

Furthermore, the lower-layer components of an OpenBTS deployment can be connected through the OpenBSC interface within a traditional GSM network architecture, rather than requiring the implementation of a full VoIP network solution (Tsou et al., 2012).

C. Asterisk

Asterisk is an open source telephony platform designed to enable diverse IP telephony hardware and software components to interconnect reliably. It operates across both Time Division Multiplexing (TDM) and packet voice layers, providing a highly flexible telephony framework. Applications supported include Interactive Voice Response (IVR), directory listing, call conferencing, call recording, and multiple-location telephony.

Asterisk is compatible with a broad range of VoIP protocols, including Session Initiation Protocol (SIP), Media Gateway Control Protocol (MGCP), H.323, and others. The platform's name derives from the '*' wildcard symbol in UNIX and DOS command-line environments, reflecting its function as a universal element within a voice network.

Asterisk has become a de facto standard for voice switching and telephony applications in the open source environment. Advances in Time Division Multiplexing and Digital Signal Processing have enabled deployment on low-cost hardware, supporting multiple packet voice protocols and delivering enterprise-grade telephony applications. Continued engagement from the open source community and forums has ensured that Asterisk remains current. The principal limitation of the platform is the relative lack of commercial investment, which has constrained the development of certain enterprise application integrations.

2.3 Benefits of Open Source Cellular Solutions

- I. **Cost:** Although not entirely without cost, the primary expenditure associated with open source telephony deployment is the acquisition of hardware devices and ancillary resources such as handsets. The Asterisk software itself is freely available. Where integration with an existing network is required, additional interface hardware may be necessary (Asterisk.org, 2014). Proprietary solutions, by contrast, typically carry licensing fees, maintenance fees, and support contract obligations, none of which apply to open source deployments.
- II. **Customisation:** Open source cellular solutions provide organisations with full administrative control over the network environment, enabling the integration of diverse applications including click-to-dial, email integration, customised digital receptionists, presence-management applications, and database connectivity. These capabilities are generally not available in hosted or commercialised solutions, where available applications are restricted to those developed by the vendor. Open source is therefore most appropriate for organisations requiring full customisation and deep integration with proprietary applications.
- III. **Interoperability:** Open source cellular solutions offer interoperability with a broad range of telephony devices from diverse manufacturers. This interoperability is built around SIP (Session Initiation Protocol), an application-layer control protocol used to initiate, modify, and terminate multimedia sessions including IP telephony calls and multicast conferences (Rosenberg et al., 2002). Further benefits include access to community-maintained support resources, availability of professional support services such as those offered by Digium for Asterisk, and the ability to add new open source applications without incurring licensing charges.

2.4 Weaknesses of Open Source Cellular Solutions

- IV. **Technical Complexity:** Open source cellular solutions demand a significant level of technical expertise. Substantial time is required to understand software codebases, configure hardware, compile software, and manage system dependencies. For example, Asterisk provides both GUI and CLI interfaces where applicable, but both require operators with appropriate technical knowledge to configure and maintain effectively.

- V. **Hardware Compatibility:** Hardware deployed in an open source environment must be verified as compatible with the software stack before implementation. This contrasts with hosted and commercial VoIP solutions, which typically ship with pre-configured servers loaded with validated software, providing a known-good deployment baseline. Any server intended to run Asterisk or an open source ISO must be evaluated for compatibility and fitness for purpose prior to deployment.
 - VI. **Security Vulnerabilities:** The open and freely accessible nature of open source cellular software introduces security risks. Contributions to the source code are made by community developers and volunteers, meaning that vulnerabilities may be introduced or persist undetected. The FBI has published advisories through the Internet Crime Complaint Center (IC3) documenting attacks in which hackers exploited security weaknesses in the Asterisk platform. Documented attack vectors include the exploitation of Asterisk as an auto-dialler, generating high volumes of phishing calls to consumers within a short period.
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3. Hosted Cellular Systems

3.1 Overview

Hosted Cellular Systems, also referred to as Hosted VoIP Systems, are telephone platforms through which calls are carried over the internet rather than over traditional copper telephone lines. The term 'hosted' indicates that the Voice over Internet Protocol service is provisioned by a commercial telecommunications provider. In this model, the service provider manages the majority of the hardware, network infrastructure, and technical resources required for system operation. Charges are typically applied on a monthly subscription basis.

This architecture enables organisations to access enterprise-grade telephony infrastructure without capital investment in physical equipment, and enables users to connect from any location including remote offices, mobile devices, and home environments while remaining integrated within the organisation's telephone system.

3.2 Benefits of Hosted Cellular Systems

- VII. **Reduced Installation Costs:** Traditional PBX system installation requires the procurement and configuration of hardware to connect the telephone service provider with IP endpoints. In a hosted cellular solution, the only hardware typically required by the client organisation is IP phones, with a suitable router in some configurations. The substantial infrastructure investment is absorbed by the service provider.
 - VIII. **Security:** The majority of infrastructure hardware in a hosted cellular solution is managed and maintained by the service provider. Services are delivered over business-grade internet connections with built-in security provisions. Multiple office locations and remote workers can therefore be connected safely and consistently without the need for client-side security management.
 - IX. **Cost Predictability:** Hosted Cellular Solutions offer cost-effective and predictable pricing for business telephony. Calls are typically charged at uniform rates regardless of the geographical distance between parties, as all calls are routed over the internet. Monthly charges are fixed and transparent.
 - X. **Feature Inclusivity:** A traditional PBX telephony system charges separately for each additional feature such as voicemail accounts, call forwarding, and conferencing. Hosted Cellular Solutions include many
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of these features within the standard service tier. Call forwarding to mobile devices when users are away from the office is typically included as a standard feature.

- XI. Ease of Administration: Hosted Cellular Solutions provide intuitive web-based and GUI administration interfaces, enabling non-technical staff to manage extensions, voicemail, and call routing. The overall management overhead is substantially lower than that of a self-hosted open source deployment.

3.3 Weaknesses of Hosted Cellular Systems

- XII. Expansion Costs: Adding extensions to an open source solution involves minimal overhead, as additional extensions are provisioned through the software configuration interface without licensing charges. In a hosted cellular solution, each additional extension typically incurs both the cost of the physical IP phone hardware and an incremental service charge, increasing the total cost of ownership as the organisation scales.
- XIII. Limited Customisation: Hosted cellular solution providers are unlikely to implement custom or bespoke features for individual business clients. The feature set available to any given organisation is generally restricted to the capabilities offered across the provider's platform. While some providers do offer customised feature tiers for specific enterprise clients, this is not universally available.
- XIV. Internet Dependency: Since all calls in a hosted cellular solution are routed over the internet, any interruption to internet service will result in a corresponding partial loss of telephony capability. While some providers offer call forwarding to voicemail or alternative numbers during outages, continuity is not guaranteed.
- XV. Voice Quality Dependency: The quality of experience in a hosted VoIP solution is directly dependent on the reliability and performance of the internet connection. Providers who oversubscribe bandwidth or compress excessive call volumes through a single pipe may deliver degraded call quality. Organisations without high-speed, reliable broadband connectivity may find that a hosted cellular solution does not perform to the required standard.

4. Comparative Summary

The table below consolidates the principal differentiating characteristics of Open Source Cellular Solutions and Hosted Cellular Systems across key operational and technical dimensions.

Feature	Open Source Cellular	Hosted Cellular
Initial Software Cost	Free (open source)	Subscription / licensing fees
Hardware Required	USRP, servers, BTS hardware	IP phones (router sometimes)
Customisation	Full administrative control	Limited to provider's feature set
Technical Expertise	High (required)	Low (managed by provider)
Security Responsibility	User / organisation	Service provider
Scalability Cost	Low (software-based extensions)	Per-unit hardware + service cost
Interoperability	Open SIP standard	Provider-dependent

Internet Dependency	Partial	Full
Voice Quality Control	Full control	Dependent on provider / bandwidth
Maintenance	Community / self-managed	Provider-managed
Best Suited For	IT-literate SMEs, research, universities	Non-technical SMEs, distributed teams

Conclusion

Open source cellular solutions offer significant cost advantages and unparalleled flexibility for technically capable organisations. Platforms such as Asterisk provide a comprehensive feature set covering conferencing, call forwarding, IVR, and more at minimal software cost. However, the technical demands of deployment, hardware compatibility verification, and ongoing maintenance represent substantial barriers to adoption for organisations without dedicated IT expertise. Adoption is therefore predominantly concentrated in IT companies, universities, and research institutions where the necessary skills are available internally.

For larger organisations, the limited availability of professional installation, maintenance, and repair services for open source cellular deployments represents a material operational risk. These organisations frequently elect to deploy hosted solutions, where installation, repair, and upgrade responsibilities are contractually assumed by the service provider. While open source may present a lower initial cost, the recurring expenditure on professional support in organisations lacking in-house expertise can make it a less economical long-term choice (Soares et al., 2008).

Hosted cellular solutions introduce their own limitations, principally around voice quality dependency on internet connection performance and the additional cost associated with scaling the number of extensions. However, the consistency, reliability, and responsiveness of provider-managed support represent compelling operational benefits for the majority of small and medium-sized enterprises.

In summary, open source cellular solutions are most appropriate for technically proficient organisations with constrained capital budgets requiring high degrees of customisation and control. Hosted cellular solutions are most appropriate for organisations prioritising operational simplicity, predictable costs, and provider-managed resilience, particularly where IT resources are limited.

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